UNIVERSITY OF BOLOGNA

Academic year 2025–2026

Advanced time series

Instructions

- You can work in groups of at most 4. Your results should be presented in a written report of at most 20 pages.
- Coursework delivery will be via Vrituale.
- Please nominate one person to upload the report and supplementary materials in Virtuale. Make sure that the names of all group members are clearly stated on the first page along with each student's ID.
- You may use any publicly available free or commercial software and packages. Indicate in your report which software and packages you have used. Also cite any other sources used in the report. Common rules for academic integrity apply.
- Include your code and data as supplementary files for reproducibility.
- This coursework is assessed and weights 50% of the exam.
- **Deadline: Monday 29th May 2026, 6:00pm (Italian time)**.

Assignment: Choose either 1 or 2

1. Choose two time series from the fMRI dataset that we have discussed in class and that you will find in Virtuale and. Further details on the dataset are also attached. Choose the two time series in such a way that one of them is non Gaussian and the other is (you can test for normality or you can detect non-Gaussianity in the diagnostic check). Fit an appropriate score-driven model and compare it with a Gaussian model. Discuss the outputs of the analysis.

2. Choose a time series over a period of your choice. You can for instance download time series from the FRED database `https://fred.stlouisfed.org/`, from the time series data library `https://pkg.yangzhuoranyang.com/tsdl/`, from the R website. In any case, document the data that you have chosen and provide the source.
   Analyse the series fitting an appropriate model score-driven model for the time varying mean or for the time varying variance. If the data are not heavy tailed, compare the results obtained with the score driven model with a corresponding Gaussian model.

Important

Please note that there is not a single correct answer and that there and the overall mark will depend on many factors such as (for example): overall quality of the report including, besides correct and rigorous procedures, quality of the writing (clarity, conciseness, formal style) and of the presentation (detailed and well explained figures, well organised paragraph structure), quality of the code (ease of implementation, detailed comments), originality (interesting, well described and well contextualised time series, critical comments and comparisons), depth of the data description.